

HUM44-3

Life Skills & Personal Finance

Your budget, your portfolio, your contract — the Finance are
turned on your own numbers

The same analytical rigour you brought to a company's accounts, now applied to the
one balance sheet you cannot delegate: your own.

Albert School — 12 hours

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Preface

In **Financial Accounting & Statements** (BUS13-3) you learned to read a company as a set of numbers: to separate what was paid from what is worth, profit from cash, a sustainable result from a one-off. You did it on a fictional firm, Lumio, and on real annual reports. This course keeps every instinct you built there and turns it through ninety degrees, onto the one entity whose statements you can never delegate to an analyst: **yourself**.

The parallels are exact, and worth holding in mind throughout. Your **budget** is your income statement — revenues in, expenses out, and a residual you choose to keep. Your **net worth** is your balance sheet — what you own minus what you owe. Your **cash flow** is the gap between the two that, exactly as for a company, can sink you even while you are nominally “profitable”. And just as a firm’s reported earnings can be distorted by accounting choices, your own financial decisions can be distorted by psychological ones — a bias this course names and disarms.

What is different is the stakes and the realism. We work only with **real** numbers — your actual rent, your actual first-job offer, the actual fees a broker charges — because the deliverables are meant to outlive the course. Across three sessions you will build three things you keep: a **budget spreadsheet** you reuse every month, a **written ETF portfolio** you can open an account and execute, and a **corrected employment contract** you can sign with your eyes open. The course is deliberately placed in Semester 4, just before your first S5 internship, so that these tools arrive precisely when the real decisions begin.

A word on scope. Tax rules, contribution rates and account types differ by country and change every year. We teach the **reasoning** — why a tax-sheltered envelope beats a taxable one over a long horizon, why a non-compete clause is priced in salary — and we anchor it in concrete French instruments (PEA, assurance-vie) with their Italian, Spanish and Swiss counterparts named alongside. Verify the current rates before you act; the logic is what transfers.

1 Part I — Foundations: the budget and the cost of time

1.1 Building a budget from real numbers

Most people’s “budget” is a vague sense that money comes in and somehow leaves. A real budget is an income statement for a person: it states, for a chosen period, exactly what comes in, exactly what goes out, and what is deliberately kept. The discipline is the same one you applied to Lumio — and so are the traps.

Worked example — a first-month budget on €1,450 net. Léa starts a paid internship in Milan at €1,450 net per month. She lists her outflows honestly, from her bank statement rather than from memory:

Inflows	€	Outflows	€
Internship (net)	1450	Rent + utilities	620
Occasional tutoring	120	Groceries	230
		Transport	55
		Phone + subscriptions	45
		Eating out / social	240
		Clothes, misc.	110
<i>Total in</i>	1570	<i>Total out</i>	1300

Residual: €270. That residual is not “what is left over” — it is the most important line on the page, and budgeting is the act of **deciding** it in advance rather than discovering it by accident at month’s end.

Definition 1 A **personal budget** is a plan, for a fixed period (usually a month), of:

- **revenues** — every euro of inflow: salary or stipend (net), side income, recurring transfers;
- **expenses** — every euro of outflow, split into **fixed** (rent, insurance, subscriptions — committed regardless of behaviour) and **variable** (groceries, going out, clothes — responsive to choices);
- a **savings target** — the residual you commit to keep, set **before** the period, not measured after it.

A budget **balances** when $\text{revenues} - \text{expenses} \geq \text{savings target}$.

The single most important move in budgeting is to treat saving as a **fixed expense paid first**, not as a leftover. This is the principle of **paying yourself first**: the savings target is transferred out on payday, and only the remainder is available to spend.

Common pitfall “I’ll save whatever is left at the end of the month.” Almost nothing is ever left, because spending expands to fill the available balance (a personal version of the soft-budget problem). Reversing the order — move the savings target out **first**, live on the rest — is the difference between people who save and people who intend to. Automate the transfer so willpower is never asked for twice.

1.1.1 What makes a savings target “defensible”

A savings target is not a wish; it must survive scrutiny. A defensible target is one you can justify on three grounds at once: it is **feasible** (the budget still balances with realistic, not heroic, variable expenses), it is **purposeful** (it funds something — an emergency buffer, then investing), and it is **prioritised** correctly.

Definition 2 The standard ordering of savings priorities, each completed before the next absorbs new money:

1. **Emergency fund** — 3 to 6 months of essential expenses, held in cash you can reach instantly. This is insurance, not investment; its job is to stop a shock (job loss, medical bill) from forcing you to sell investments or borrow at punitive rates.
2. **High-interest debt** — any consumer debt or card balance above roughly 8–10% is repaid before investing, because **paying off** a 15% debt is a guaranteed 15% return, which no portfolio can promise.
3. **Long-term investing** — only once the buffer exists and expensive debt is gone does new saving flow to the portfolio of Part II.

Worked example — turning Léa's residual into a target. Léa's €270 residual is real but fragile. She has no emergency fund and a €600 phone-financing balance at 0% (so not “expensive debt”). She sets her target at €220/month — feasible (it leaves a €50 cushion against a bad month), purposeful (directed first at a €4,000 emergency fund $\approx$ 4 months of essentials), and prioritised (buffer before portfolio). She automates a €220 transfer on the 25th. The €4,000 buffer is reached in about eighteen months; after that the €220 redirects to investing.

The deliverable for this session is a **spreadsheet** — not a paper list — precisely because a budget is a living model you re-run each month. A good spreadsheet has one row per category, columns for **budgeted** versus **actual**, and a formula that flags any category where actual overshoots budget. The point is not the first month's plan; it is the monthly **variance review** that turns the plan into a habit.

Common pitfall **Forgetting the irregular expenses.** The dangerous costs are not the monthly ones — those you feel. They are the annual or surprise outflows: insurance premiums, a flight home, a deposit, an unexpected tax balance. Budget them by taking the yearly total, dividing by twelve, and setting that twelfth aside each month into a “sinking fund”. A budget that ignores irregular expenses looks balanced and is not.

1.2 Compound interest: the cost of starting late

Ask why an emergency fund comes before investing and the answer is **risk**. Ask why investing should start as early as humanly possible and the answer is a single piece of arithmetic that most people underestimate by an order of magnitude: compound interest over a long horizon. We make the cost of delay **quantitative** rather than rhetorical.

Worked example — the ten-year head start. Two students each save €200 a month into a portfolio returning 7% a year. Aïcha starts at 25 and stops contributing at 65 — 40 years. Bruno starts at 35 — 30 years. Aïcha contributes €96,000 of her own money over her life;

Bruno contributes €72,000. The difference in contributions is €24,000. The difference in final value is far larger:

$$\text{Aïcha} \approx \text{€}525\{\text{, } \}000, \quad \text{Bruno} \approx \text{€}244\{\text{, } \}000.$$

A ten-year head start — €24,000 of extra contributions — produces roughly **€281,000** of extra wealth. The gap is not the money saved; it is the **years of compounding** that the early money buys.

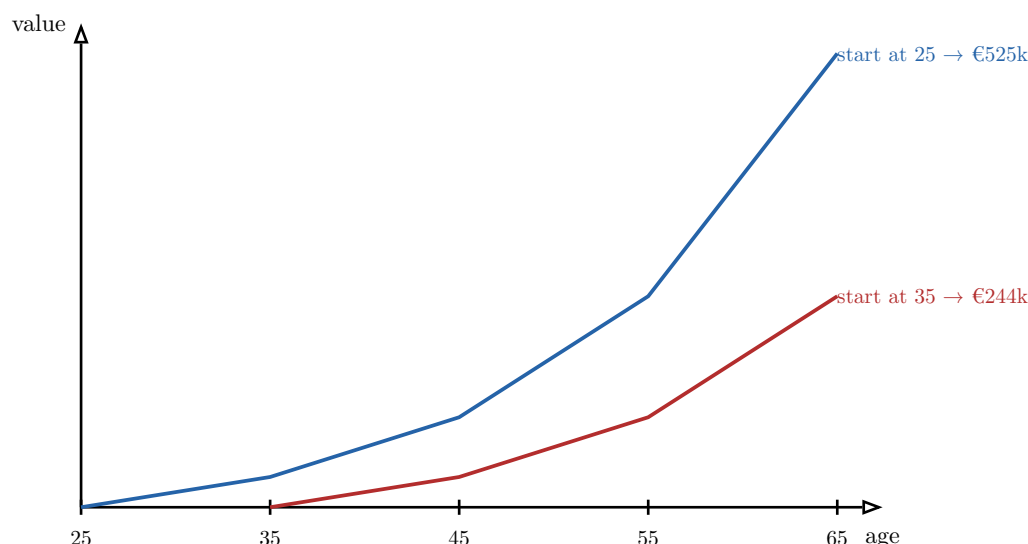


Figure 1: Two identical €200/month savers at 7%. The only difference is a ten-year head start, yet the early starter ends with more than double the wealth — because compounding is exponential, the last decades do the heaviest lifting, and only the early starter has them.

Definition 3 Compound interest is interest earned on both the original principal **and** on the interest already accumulated. A sum P growing at annual rate r for n years becomes

$$P(1 + r)^n.$$

A stream of regular contributions C per period at periodic rate r over n periods (a **future value of an annuity**) grows to

$$C \cdot \frac{(1 + r)^n - 1}{r}.$$

The exponent n is what makes time, not the contribution size, the dominant lever over a long horizon.

Remark A useful piece of mental arithmetic is the **Rule of 72**: money compounding at $r\%$ per year roughly **doubles** every $72/r$ years. At 7%, that is about every ten years. So €10,000 invested at 25 becomes €20k by 35, €40k by 45, €80k by 55, €160k by 65 — four doublings, from a single early decision.

Common pitfall **Confusing nominal with real returns.** A 7% return with 2% inflation is only about 5% in **purchasing power**. Over 40 years inflation is not a footnote: at 2%,

prices roughly double, so a “€525,000” projection buys what €240,000 buys today. Always know whether a return is nominal (before inflation) or real (after). When comparing to a savings goal expressed in today’s money, use real returns.

Common pitfall Compounding cuts both ways. The same arithmetic that builds wealth on a 7% portfolio **destroys** it on an 18% credit-card balance — which is exactly why expensive debt is repaid before investing. A debt compounding against you at 18% doubles every four years.

2 Part II — Investing: from asset classes to a written portfolio

2.1 The asset classes and their risk–return logic

Before choosing **what** to buy, you must understand the menu. Every investment belongs to an **asset class**, and each class occupies a characteristic position on the trade-off between risk and expected return. There is no asset that is high return and low risk; if there were, everyone would hold only it and its price would rise until the return fell. This is the foundational fact of investing.

Definition 4 An **asset class** is a group of investments with similar economic behaviour and risk drivers. The four core classes:

- **Cash and equivalents** — bank deposits, money-market funds. Essentially no risk of nominal loss; lowest return; loses to inflation over time. Horizon: months.
- **Bonds** (fixed income) — loans to governments or companies that pay interest and return principal at maturity. Moderate risk (the borrower may default; prices fall when rates rise); moderate return. Horizon: a few years.
- **Equities** (stocks) — ownership shares in companies. High short-term volatility; highest long-run expected return; can fall 30–50% in a crisis. Horizon: a decade or more.
- **Real estate** — property held directly or via listed vehicles (REITs). Return between bonds and equities; illiquid if held directly; sensitive to interest rates. Horizon: many years.

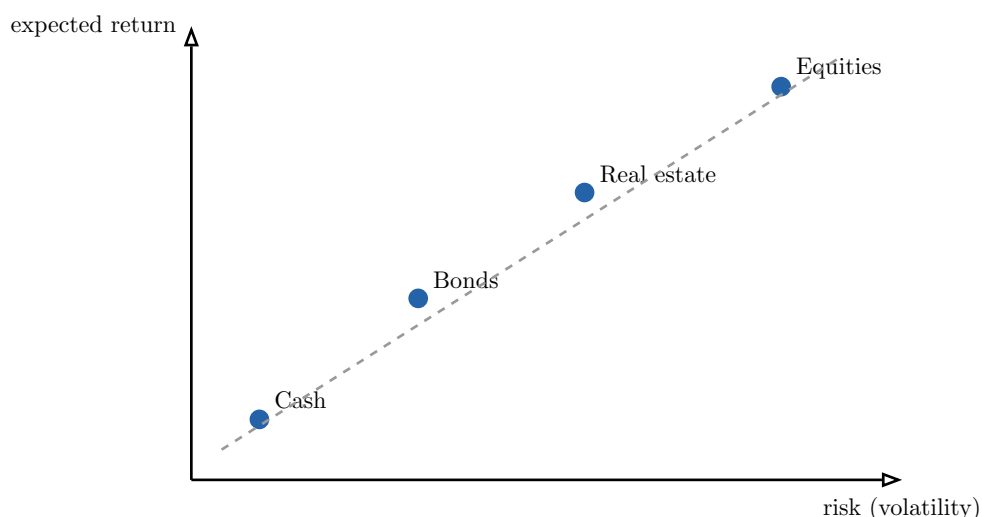


Figure 2: The risk–return spectrum. Higher expected return is only available by accepting higher risk (here, volatility). The dashed line is the trade-off itself: there is no free lunch above it. Which class suits you depends on your **time horizon** — the longer you can leave money untouched, the more volatility you can afford to ride out.

The decisive variable in matching a class to a goal is **time horizon** — how long before you need the money. Volatility is only a danger if you are forced to sell during a dip. Equities have fallen by half and recovered many times; over a single year that is terrifying, but over twenty years it has historically been the highest-returning class. Cash is the opposite: safe over a year, but a guaranteed slow loss over twenty once inflation is counted.

Common pitfall Matching the wrong horizon to the asset. Two symmetric, costly errors: holding your house deposit (needed in two years) in equities — where a crash right before you buy could wipe out 30% — and holding your 40-year retirement savings in cash — where inflation quietly erodes it. The rule: **short-horizon money in safe assets, long-horizon money in growth assets.** The emergency fund of Part I is short-horizon by definition, which is why it stays in cash.

2.2 Tax envelopes: keeping more of what you earn

Two investors buy the identical portfolio and earn the identical return. One ends up meaningfully richer — because she held it inside a **tax-advantaged envelope** and he held it in a plain taxable account. An envelope is not an investment; it is a legal **wrapper** around investments that changes how gains are taxed. Over a 40-year horizon, the tax drag avoided is one of the largest controllable levers on final wealth.

Worked example — the drag of annual taxation. Suppose a portfolio grows 7% a year and gains are taxed at 30%. In a **taxable** account where you are taxed on gains as they arise, your effective compounding rate falls toward 5%. In a **sheltered** envelope where gains compound untaxed until withdrawal, you compound at the full 7% and pay tax once, at the end. Over decades the gap between compounding at 7% and at 5% is enormous — the same exponential effect that rewarded the early starter now rewards the sheltered one.

Definition 5 A **tax envelope** is an account type with special tax treatment, usually offered in exchange for accepting constraints (a holding period, a contribution ceiling, limits on what

may be held). The reasoning for choosing one rests on two axes: **time horizon** (how long until you withdraw — longer favours envelopes that reward patience) and **tax treatment** (what is taxed, when, and at what rate).

Definition 6 Principal envelopes by country of residence — names differ, logic rhymes:

- **France.** The **PEA** (Plan d'Épargne en Actions): holds European equities/funds; gains become income-tax-free after 5 years (social charges still apply); capped contributions. The **assurance-vie**: a flexible wrapper holding funds and bonds; favourable taxation and inheritance treatment after 8 years.
- **Italy.** The **PIR** (Piano Individuale di Risparmio): exempts gains from capital gains tax if held 5 years, with investment-composition constraints.
- **Spain.** The **Plan de Pensiones** (pension plan, contributions deductible, locked until retirement) and **fondos de inversión** benefiting from tax-deferred switching between funds.
- **Switzerland. Pillar 3a** (pension 3e pilier): contributions deductible up to an annual ceiling, locked until retirement, taxed lightly on withdrawal.

Common pitfall **Letting the tax tail wag the investment dog.** An envelope's tax break is worth nothing if its constraints force a bad portfolio. A Swiss Pillar 3a locks money until retirement — excellent for retirement saving, useless for a deposit you need in three years. Choose the **goal and horizon first**, then the envelope that fits, then the investments inside it — never the reverse.

Remark Envelopes are not mutually exclusive and rarely an all-or-nothing choice. A common structure: long-horizon equity growth inside a PEA (or local equivalent), more flexible medium-term money in an assurance-vie, and the emergency fund outside any envelope in plain cash where you can reach it instantly.

2.3 Constructing your first portfolio

Now we assemble the deliverable: a **written** first portfolio you could open an account and execute tomorrow, with €500 to €5,000, under the real fees and minimums a retail platform imposes. Every choice below is defended on evidence, because the second learning outcome of this course is that you can **justify** the portfolio, not merely state it.

2.3.1 Why index funds, not stock-picking

Definition 7 An **index fund** (commonly an **ETF**, exchange-traded fund) holds every security in a market index in proportion to its weight, charging a very low fee. An **active fund** employs managers who try to beat the index by selecting securities, charging a much higher fee.

Worked example — the evidence on active vs passive. The question is empirical, and the data are stark: over any 10–15 year window, the large majority of active equity funds **underperform** their benchmark index after fees — and the few that win in one period are mostly not the winners of the next. The mechanism is simple arithmetic, not luck: all investors

together **are** the market, so before costs the average actively managed euro earns the market return; after the higher fees of active management, it must on average earn **less**. A 1.5% annual fee versus a 0.2% ETF fee is, over 40 years of compounding, a difference of roughly a third of your final wealth.

Common pitfall Confusing past performance with skill. A fund advertising “+22% last year!” tells you almost nothing about next year. Markets are close enough to efficient that a single year’s winner is usually explained by chance or by a risk that has not yet bitten. Regulators force the disclaimer “past performance is not indicative of future results” precisely because investors so reliably ignore it.

2.3.2 Diversification and the three building blocks

Definition 8 Diversification is holding many uncorrelated assets so that the idiosyncratic failure of any one does little damage to the whole. It is the closest thing to a free lunch in finance: it reduces risk **without** reducing expected return, because the random ups and downs of individual holdings partly cancel.

A globally diversified beginner portfolio can be built from three low-cost ETF blocks:

Definition 9 The three building blocks:

- **Global equities** — a **MSCI World** (developed markets) or all-world ETF, holding 1,500+ companies across dozens of countries in one line. This is the growth engine.
- **Bonds** — a broad government/investment-grade bond ETF, the ballast that cushions equity crashes.
- **Listed real estate** — a **REIT** ETF, adding a class that does not move in lockstep with stocks or bonds.

The **split between them** is the single most important decision and is driven by your risk profile and horizon: more equities for a long horizon and a high tolerance for swings, more bonds for a short horizon or low tolerance.

Worked example — a written €3,000 portfolio for a 23-year-old. Profile: long horizon (decades), comfortable with volatility, no need for this money soon. Target allocation 80% equities / 10% bonds / 10% real estate:

Holding	Target	€	Fee (TER)
MSCI World ETF (accumulating)	80%	2,400	0.20%
Global aggregate bond ETF	10%	300	0.10%
Developed-markets REIT ETF	10%	300	0.30%
Total	100%	3,000	<i>≈ 0.19% blended</i>

Justification (the deliverable). Equity-heavy because the horizon is long enough to ride out crashes; accumulating share class so dividends reinvest automatically inside the envelope; a small bond/REIT sleeve for diversification; total fee 0.19% versus 1.5% for an active equivalent — the difference compounds in the investor’s favour for forty years. Held inside a PEA-style envelope where available.

2.3.3 Two habits that make it work: DCA and rebalancing

Definition 10 Dollar-cost averaging (DCA): investing a **fixed amount on a fixed schedule** (e.g. €250 on the 1st of each month) regardless of price. You automatically buy more units when prices are low and fewer when high, removing the impossible task of timing the market and the emotional pull to buy at peaks and sell at troughs.

Worked example — DCA buys more when it's cheap. Investing €240 each month into an ETF whose price swings around:

Month	Price	Units bought
Jan	€120	2.0
Feb	€80	3.0
Mar	€60	4.0
Apr	€96	2.5

Total invested €960, total units 11.5, **average price paid** €83.5 — below the €89 simple average of the four prices, because the fixed euro amount bought more units in the cheap months. DCA mechanically tilts your average cost downward.

Definition 11 Rebalancing: periodically (e.g. once a year) selling whatever has grown beyond its target weight and buying whatever has fallen below, to restore the original allocation. After a strong equity year the portfolio drifts to, say, 88% equities; rebalancing trims it back to 80%. It enforces “sell high, buy low” as a rule rather than a decision, and keeps risk from creeping up unnoticed.

Common pitfall Over-trading and fee leakage. The opposite danger to inaction. Checking the portfolio daily, chasing the latest hot ETF, rebalancing monthly — each trade costs a fee and a spread, and frequent tinkering reliably **lowers** returns. Annual rebalancing and monthly automated DCA are enough. A first portfolio should be, deliberately, boring.

2.4 Investor psychology: the enemy is usually you

The portfolio above is simple to write and hard to hold — not because the finance is difficult, but because the **psychology** is. The largest destroyer of private investors' returns is not bad assets; it is predictable behavioural errors made under stress. Naming them is the first step to neutralising them.

Definition 12 Three biases that most damage investors:

- **Loss aversion** — a loss hurts about twice as much as an equal gain feels good. It makes investors sell in a crash to stop the pain, locking in the loss at the worst moment, and hold losers too long hoping to “get back to even”.
- **FOMO** (fear of missing out) — the urge to pile into whatever has risen spectacularly (a hot stock, a crypto, a meme), buying at the top precisely because it has already gone up.
- **Overconfidence** — overestimating one's ability to pick winners or time the market, leading to under-diversification and over-trading.

Worked example — loss aversion in a crash. In a 35% market crash, an investor with our €3,000 portfolio sees it fall to €2,000. Loss aversion screams to “stop the bleeding” and sell to cash. But selling converts a **paper** loss into a **realised** one and guarantees missing the recovery — which historically has followed every crash. The investor who did nothing (or kept buying via DCA) recovered and went on; the one who sold at the bottom did the single most expensive thing available.

Definition 13 The practical **rules designed to neutralise** these biases — note that each is a rule fixed in advance, so that the decision is made when calm, not in the moment:

1. **Automate** contributions (DCA) so buying never depends on how you feel.
2. Write an **Investment Policy Statement**: a one-page note stating your target allocation and the rule “I will not sell in a downturn; I will rebalance once a year.” Re-read it during crashes.
3. **Don’t check the portfolio often** — quarterly is plenty. Frequent checking amplifies loss aversion via the daily noise.
4. **Diversify and index** — structurally removes the overconfident bet on a single winner.
5. Impose a **cooling-off rule**: no trade acted on the day you first want to make it.

Common pitfall **Believing you are the exception.** Every investor reading the list above quietly assumes the biases apply to **other** people. That belief is itself overconfidence — the most dangerous bias, because it disables the defences against the others. The professionals who survive market panics are not those who feel no fear; they are those who pre-committed to rules and removed their in-the-moment selves from the decision.

3 Part III — Rights: payslip, contract, and protection

3.1 Decoding a payslip line by line

You negotiate a salary as a single number, but you receive something quite different, and the gap is large and instructive. A payslip is the income statement of your employment: it starts from the total cost you generate for your employer and subtracts, in a deliberate order, the contributions and taxes that stand between that cost and the money in your account. Reading it line by line is the first step in **total-compensation** reasoning.

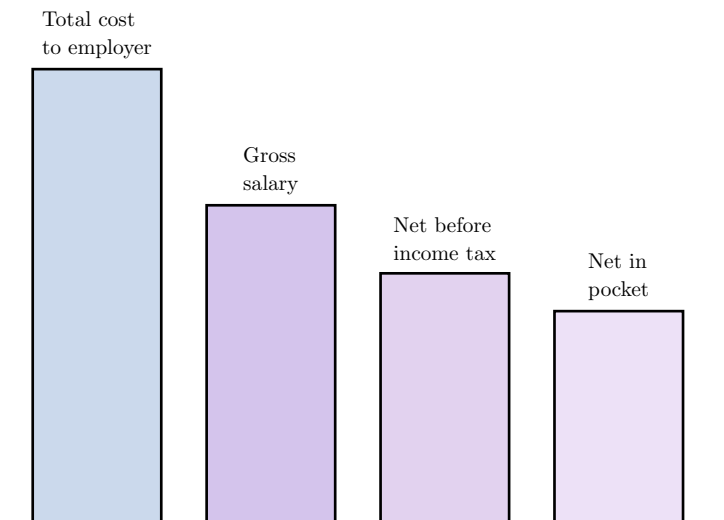


Figure 3: The payslip cascade. The employer’s true outlay (super-gross) exceeds your gross by **employer** contributions; gross exceeds net by **employee** contributions; net pay exceeds take-home only after **income tax**. The headline salary you negotiated is the second bar, not the first or last. Heights are illustrative.

Definition 14 Reading down a payslip:

- **Gross salary** (salaire brut) — the figure in your contract, before any deduction.
- **Employer contributions** (cotisations patronales) — paid by the employer **on top** of your gross, funding health, pension, unemployment, etc. They do not appear in your gross but are part of your true cost (gross + employer contributions = **super-gross**, the employer’s total outlay).
- **Employee contributions** (cotisations salariales) — deducted **from** your gross, funding the same social systems from your side.
- **Net taxable income** — gross minus employee contributions; the base on which income tax is computed.
- **Income tax** (in France, withheld at source — prélèvement à la source).
- **Net pay** (net à payer) — what actually lands in your bank account.

Worked example — from €40k cost to take-home. An employer budgets €40,000 a year to employ you. Roughly (rates vary by country and bracket):

Total cost to employer (super-gross)	40,000
– Employer contributions (28%)	(8,750)
Gross salary	31,250
– Employee contributions (22%)	(6,875)
Net taxable income	24,375
– Income tax (illustrative)	(2,200)
Net in pocket	≈ 22,175

The employer spends €40,000; you receive €22,000. The €18,000 difference is not lost — it buys healthcare, pension rights and unemployment cover — but you must **see** it to reason about your real compensation.

Definition 15 **Total compensation** is the full economic value of a job, not just net pay: gross salary **plus** the value of employer contributions (pension and health rights you accrue), **plus** benefits (bonus, equity, meal vouchers, insurance, training). Two offers with the same gross can differ sharply in total compensation.

Common pitfall Comparing offers on gross — or worse, on net — alone. A €34k offer with strong employer pension contributions, health cover and a bonus can be worth more than a €37k offer with none. And comparing a **net** figure across countries is meaningless without knowing what the contributions buy: a higher net in a country where you must privately fund health and pension may leave you worse off. Compare **total compensation**, and know what the deductions purchase.

3.2 The employment contract and its five decisive clauses

An offer letter is a number; the **contract** is where the number's conditions live. Most of a contract is boilerplate, but five clauses materially shape your freedom, your finances and your future, and they are the ones to read — and negotiate — before signing.

Definition 16 The **five decisive clauses** in almost any employment offer:

1. **Non-compete** (clause de non-concurrence) — restricts where you may work after leaving (which competitors, which geography, for how long). In many jurisdictions it is enforceable **only if** limited in time and space and **financially compensated**; an uncompensated or unlimited non-compete is often void.
2. **Confidentiality** (NDA) — bars you from disclosing the employer's secrets; typically survives the end of employment and is broadly enforceable.
3. **IP assignment** — assigns to the employer the intellectual property you create (see the next chapter); the single clause most worth scrutinising for anyone who builds, writes or codes.
4. **Mobility** (clause de mobilité) — lets the employer change your work location within a defined zone; an open-ended mobility clause can mean an unwanted move.
5. **Notice period** (préavis) — how much warning either side must give to end the contract; it cuts both ways (your security, but also your inability to leave quickly for a better offer).

Worked example — what a clause is really worth. An offer carries a 12-month non-compete covering all of Europe, **uncompensated**. Read literally, it would bar the signatory from the entire industry for a year with no pay — and in several jurisdictions is therefore unenforceable as written. The right move is not to sign and hope, nor to refuse outright, but to **negotiate**: narrow it to direct competitors, shorten it to 6 months, and attach the legally required compensation. A clause understood is a clause you can price and bargain.

Common pitfall “It’s standard, just sign it.” Boilerplate is still binding. The pressure to sign quickly — a deadline, an eager recruiter — is exactly when costly clauses slip through. Read all five clauses, ask what each means in your worst case, and remember that **everything in an offer is negotiable before signature and almost nothing after**. Get changes in writing, in the contract — not in a reassuring email.

3.3 Intellectual property at work

For anyone in a technical or creative role, the IP-assignment clause deserves its own treatment, because the **default** rules — what happens if the contract is silent — are often counter-intuitive and vary by country and by the **type** of creation.

Definition 17 Typical **default** ownership rules (absent a contrary clause):

- **Inventions made in the course of employment**, using the employer’s resources and within your assigned mission, generally belong to the **employer** — though many countries grant the employee inventor additional compensation for a patented “mission invention”.
- **Copyright** in works (code, text, design) created as an employee is, in many systems, assigned to or owned by the employer for works made on the job — but the rules differ sharply between, e.g., common-law “work made for hire” and French author’s-right traditions, where assignment must often be **explicit**.
- Work created **outside** your mission, on your own time and resources, is more likely to remain **yours** — but a broadly drafted clause may try to claim it.

Worked example — the side-project trap. An engineer builds a small app on weekends, unrelated to her job. Her contract contains a sweeping clause assigning to the employer “all intellectual property created during the period of employment.” Read literally, that could capture her side project. The fix is to **renegotiate the scope before signing**: limit assignment to work within her mission and using company resources, and, where she has existing or planned side projects, list them explicitly as carve-outs (a “prior inventions” schedule).

Common pitfall Assuming “my own time, my own project” protects you. The default may favour you, but an over-broad contractual clause can override the default — and you signed it. If you have, or might start, anything of your own, raise it **before** signing and get the carve-out in writing.

3.4 Social protection and tax basics across job changes

A job is not only a salary; it is a bundle of **social protections** — health cover, disability insurance, supplementary pension — much of it tied to the employer. Changing jobs reshuffles that bundle, and some of it is lost silently unless you act at the moment of signing. The final skill of this course is to map what is preserved and what is lost across a transition, and to handle the tax basics that come with a first job.

Definition 18 The main social protections to track across an employer change:

- **Health insurance** — public/statutory cover usually continues, but the **supplementary** (employer “mutuelle”/private top-up) typically **ends** with the job; check for portability rights and arrange a bridge so you are never uncovered.
- **Disability and death cover** (prévoyance) — often employer-provided and **lost** on leaving; a gap here is the most dangerous because the events it covers are rare but catastrophic.
- **Supplementary / occupational pension** — contributions stop; accrued rights are usually preserved but may need to be **transferred** or kept track of, or they are forgotten.
- **Unemployment rights** — depend on how the contract ends (resignation vs termination vs mutual agreement), which can determine whether you draw benefits at all.

Worked example — the gap nobody mentions. Maya leaves Job A on the 30th and starts Job B on the 15th of next month — a two-week gap. Her statutory health cover continues, but Job A's supplementary health and her disability cover **ended** on the 30th, and Job B's start on the 15th. For two weeks she is exposed on exactly the risks (accident, serious illness) that matter most. The decisions to make **at signing**: confirm Job B's protections' start date, arrange portability or a short private bridge, and transfer or note the accrued pension rights from A.

Definition 19 First-job **tax-declaration** basics:

- You are generally required to **file an annual tax return** in your country of residence even if tax was withheld at source, to reconcile what was withheld against what is owed.
- Know your **residence for tax** — it usually determines where you declare your worldwide income; an internship abroad can complicate this.
- Keep the **documents**: your annual salary statement, proof of any deductible expenses, and records of investment income (which your envelopes from Part II may shelter or simplify).

Common pitfall Assuming “tax is withheld, so I don't need to do anything.” Withholding at source is an **estimate**; the annual return reconciles it, and skipping it can forfeit a refund or trigger penalties for an underpayment. And the decisions that matter most — the protection bridges, the pension transfer — must be made **at the job change**, not discovered months later when the gap has already cost you.

The thread of this course. You brought a company's statements under analytical control in BUS13-3; here you did the same for your own life. A **budget** is your income statement, with saving paid first as a fixed expense. **Compound interest** makes time the dominant lever, so you start early and shelter gains in the right **tax envelope**. A first **portfolio** is global, indexed, cheap, diversified, fed by automatic **DCA** and kept on course by annual **rebalancing** — and held steady against your own **loss aversion, FOMO and overconfidence** by rules fixed in advance. On the income side, you read a **payslip** from the employer's true cost down to your take-home and reason in **total compensation**; you read a **contract** for its five decisive clauses; you guard your **IP** with explicit carve-outs; and you map what is preserved and lost in **social protection** at every job change. Three deliverables — a budget, a written portfolio, a corrected contract — outlive the course and arrive exactly when the real decisions begin.